I, Calculations.

Bài 1:

Cho $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$; $W_q = W_k = W_v = I_2$

$$QK^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Vì $d_k = 2$

$$S = \frac{QK^T}{\sqrt{2}} \approx \begin{bmatrix} 0,7071 & 0 & 0,7071 \\ 0 & 0,7071 & 0,7071 \\ 0,7071 & 0,7071 & 1,4142 \end{bmatrix}$$

Softmax theo từng hàng

$$A \approx \begin{bmatrix} 0,4011 & 0,1978 & 0,404 \\ 0,1978 & 0,404 & 0,4011 \\ 0,2483 & 0,2483 & 0,5035 \end{bmatrix}$$

Context vector: $Z = AV \approx \begin{bmatrix} 0,8022 & 0,5989 \\ 0,5989 & 0,8022 \\ 0,7517 & 0,7517 \end{bmatrix}$

Bài 2:

$head_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$; $head_2 = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$

$$\text{concat}(head_1, head_2) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 1 & 1 & 0 & 2 \end{bmatrix}_{3\times 4}$$

Chọn $W_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow \text{Concat}(head_1, head_2).W_0 = \begin{bmatrix} 3 & 1 \\ 3 & 2 \\ 2 & 4 \end{bmatrix}$

Bài 3:

$PE(pos) = [\sin(pos), \cos(pos), \sin(pos/100), \cos(pos/100)]$

$PE(0) = [0, 1, 0, 1]$

$PE(1) \approx [0.8415; 0.5403; 0.01; 0.99995]$

$PE(2) \approx [0.9093; -0.4161; 0.02; 0.99980]$

Bài 4:

Cho $x = [1, 2]^T$, $W_1 = I_2$; $b_1 = [0, -1]^T$

$h = Relu(W_1 x + b_1) = Relu([1, 1]^T) = [1, 1]^T$

~~Đề bài~~ $W_2 = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$; $b_2 = [1, -1]^T$

$y = W_2 h + b_2 = [4, 3]^T$

Bài 5:

Với $R = 8$, $T = 100$, $d_{model} = 128$, $n_{heads} = 4$

$\therefore d_k = 128/4 = 32$

Bài 6:

T	kích thước Matrix	Số phần tử
10	10×10	100
100	100×100	10000
500	500×500	250000
1000	1000×1000	1000000
10000	10000×10000	100000000

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